

Markus Walker

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Enterprise AI Engineering and Delivery | Secure AI | Anthropic Claude Partner Network certified | AGSVA Baseline, eligible for NV1 and NV2

Summary

AI engineer with 13 years in IT infrastructure, including seven years across IT and operational technology (OT) operations and the last two years focused on AI engineering and Secure AI. Delivers secure AI solutions end to end, from use-case assessment and architecture through build, controls and handover. Designs secure agentic AI architectures across Azure, AWS and Anthropic platforms, and has a deep interest in knowledge bases for AI agents, from knowledge graphs and graph ontologies to vector retrieval. Builds so the AI model can be swapped: open knowledge formats, standard protocols and instructions any agent can read.

Skills

AI engineering: Generative AI, large language models (LLMs), AI agents, multi-agent design, agentic workflows, prompt engineering, Model Context Protocol (MCP) servers, custom plugins, n8n, Claude, Claude Code, Codex and ChatGPT. Secure AI and cyber security: AI threat modelling, AI security capability assessment, prompt injection testing, secure software development lifecycle (SDLC), AI governance, MCP governance, cloud security, identity and access management (IAM) and incident response planning. Knowledge engineering: Knowledge bases and knowledge graphs, graph ontologies, graph databases such as Neo4j, vector embeddings and semantic retrieval, Open Knowledge Format (OKF), ingestion pipelines, source provenance and validation. Software engineering: Python (FastAPI, Pydantic), TypeScript, React, Playwright, pytest, PowerShell, Git and GitHub. Cloud and platforms: Microsoft Azure (Azure AI Foundry, Azure OpenAI, Azure Virtual Desktop), AWS (Bedrock, Bedrock AgentCore), Oracle Cloud Infrastructure, Microsoft 365, Entra ID, Intune, Windows Autopilot, SCCM and GitHub Enterprise. Consulting and delivery: Client presentations and workshops, AI use-case and architecture assessment, white papers, stakeholder engagement, enterprise networking and change control.

Experience

Senior Engineer, Secure AI | CyberCX, part of Accenture | Jun 2026 to Present

- Delivered secure AI solutions for clients end to end, from use-case assessment and solution architecture through build, security controls and handover.
- Deployed sovereign AI environments on Azure: virtual machines built from golden images, Azure AI Foundry with Azure OpenAI models hosted in an Australian region, and mobile device management (MDM) and access controls that restrict each machine to the approved models.
- Delivered client presentations and workshops on frontier AI and security, including the Claude Mythos briefing, and advised on MCP governance, secure SDLC and enterprise AI adoption.
- Built custom AI plugins and agentic workflows, with versioned distribution through a GitHub Enterprise marketplace.
- Built knowledge bases for people and AI agents, and AI-assisted document-generation workflows with reusable skills, agents and review processes.
- Author of Distinguishing AI-Driven Attacks from Human and Automated Activity, published by CyberCX, translating research into security operations centre (SOC) guidance for detection, triage and response.

Independent AI Practice and Professional Development | 2025 to 2026

- AI security research and certification study, with projects in AWS security architecture, incident response, Splunk and Active Directory security labs.

IT Field Engineer | Tata Consultancy Services | May 2019 to Aug 2025

Embedded contractor running IT and OT for Shell QGC and QCLNG upstream and midstream operations in Queensland, the same assets built and commissioned in earlier site roles.

- Delivered IT field engineering across 20+ remote operational sites, including camps, plants and drilling environments, covering planned delivery, escalations and after-hours support.
- Maintained business-critical Microsoft 365, networking and telecom platforms supporting control-room operations and service continuity.
- Executed network transformation, including 300+ Cisco to Aruba access point replacements, and delivered Starlink connectivity for remote operations from project development through deployment.
- Administered Entra ID identity and access management, including role-based access control and device compliance, and supported the move from SCCM to Intune, Windows Autopilot and Azure AD Join.

Junior Network Engineer | Data#3 | Jan 2019 to May 2019

- Field engineering and cutover coordination for the Shell QGC Enterprise Network Transformation project.

Earlier roles | 2005 to 2019

QGC site roles with MSS, ECM and Protech (2012 to 2019): worked on gas plant projects from breaking ground, including laying the copper earth grid, through construction and commissioning to handover into operations. Lighting and Sound Engineer, Airco Audio, USA (2005 to 2012): tertiary-trained in lighting and sound engineering, running networked lighting and audio control systems in live production.

Projects

Secure Agentic AI Architecture

- Designed a secure agentic AI reference architecture that keeps the logical design independent of any one cloud or model, and an AI security capability matrix that compares major AI platforms against common control components.

Knowledge Bases for AI Agents

- Designs and builds knowledge bases that AI agents can reason over, from graph ontologies and vector-searchable knowledge graphs to open-format repositories, with ingestion, redaction, retrieval and provenance tooling.

MCP and Agent Tooling

- Built MCP servers, an MCP configuration security auditor, self-hosted AI agent services and React and TypeScript applications.

Certifications and Education

Anthropic Claude Partner Network certified: Claude Certified Associate, Foundations | 2026

AWS Certified Solutions Architect, Associate | 2025

Oracle Cloud Infrastructure (OCI), three certifications including Generative AI Professional | 2025

Certificate IV in Cyber Security (22603VIC) | TAFE Queensland | Mar 2026

University studies in audio engineering | St. Louis, USA